

1.

Let the conditional densities for a two-category one-dimensional problem be given by the Cauchy distribution

$$p(x|\omega_i) = \frac{1}{\pi b} \cdot \frac{1}{1 + \left(\frac{x-a_i}{b}\right)^2}$$

with $i = 1, 2$.

(a)

If $P(\omega_1) = P(\omega_2)$, show that $P(\omega_1|x) = P(\omega_2|x)$ if $x = (1/2)(a_1 + a_2)$. Sketch $P(\omega_1|x)$ for the case $a_1 = 3, a_2 = 2, b = 5$. How does $P(\omega_1|x)$ behave as $x \rightarrow -\infty$? as $x \rightarrow \infty$?

We essentially aim to show that if the prior probabilities are equal, then the posterior probabilities are also equal at the midpoint in between the two means a_1 and a_2 . We apply Bayes' theorem to express the posterior probabilities:

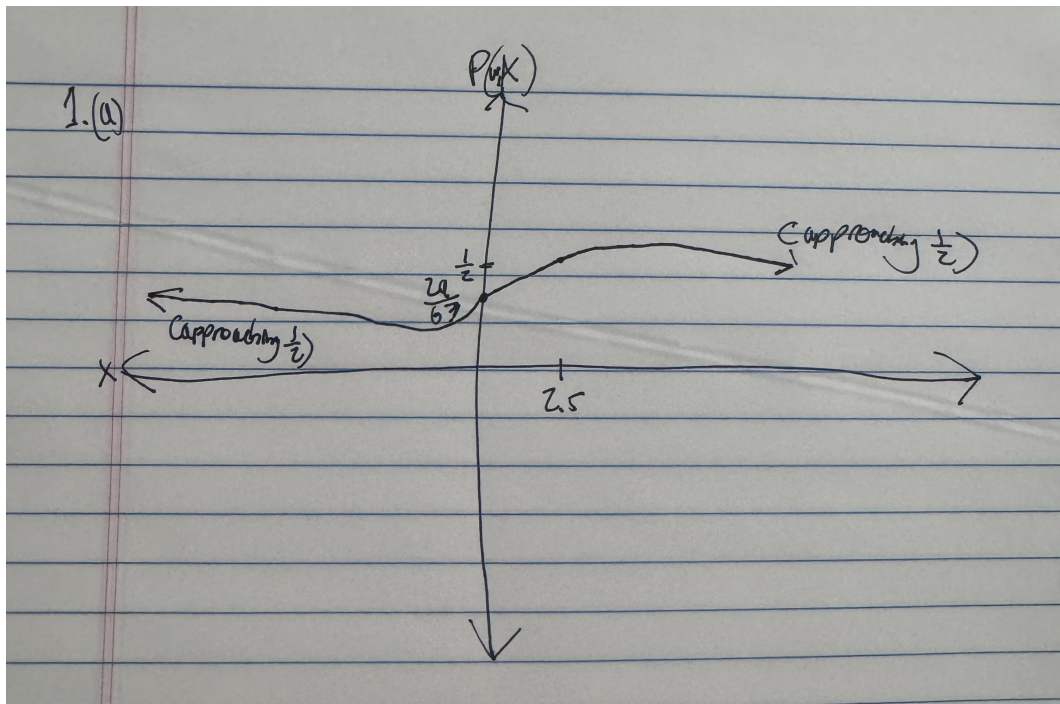
$$\begin{aligned} P(\omega_1|x) &= \frac{p(x|\omega_1)P(\omega_1)}{p(x)} \\ &= \frac{p(x|\omega_1)P(\omega_1)}{p(x|\omega_1)P(\omega_1) + p(x|\omega_2)P(\omega_2)} \\ &= \frac{p(x|\omega_1) \cdot \frac{1}{2}}{p(x|\omega_1) \cdot \frac{1}{2} + p(x|\omega_2) \cdot \frac{1}{2}} \\ &= \frac{p(x|\omega_1)}{p(x|\omega_1) + p(x|\omega_2)} \\ &= \frac{\frac{1}{\pi b} \cdot \frac{1}{1 + \left(\frac{x-a_1}{b}\right)^2}}{\frac{1}{\pi b} \cdot \frac{1}{1 + \left(\frac{x-a_1}{b}\right)^2} + \frac{1}{\pi b} \cdot \frac{1}{1 + \left(\frac{x-a_2}{b}\right)^2}} \\ &= \frac{\frac{1}{1 + \left(\frac{x-a_1}{b}\right)^2}}{\frac{1}{1 + \left(\frac{x-a_1}{b}\right)^2} + \frac{1}{1 + \left(\frac{x-a_2}{b}\right)^2}} \\ &= \frac{\frac{1}{\frac{b^2 + (x-a_1)^2}{b^2}}}{\frac{1}{\frac{b^2 + (x-a_1)^2}{b^2}} + \frac{1}{\frac{b^2 + (x-a_2)^2}{b^2}}} \end{aligned}$$

$$\begin{aligned}
&= \frac{\frac{b^2}{b^2+(x-a_1)^2}}{\frac{b^2}{b^2+(x-a_1)^2} + \frac{b^2}{b^2+(x-a_2)^2}} \\
&= \frac{\frac{1}{b^2+(x-a_1)^2}}{\frac{1}{b^2+(x-a_1)^2} + \frac{1}{b^2+(x-a_2)^2}} \\
&= \frac{\frac{1}{b^2+(x-a_1)^2} \cdot (b^2 + (x-a_1)^2) \cdot (b^2 + (x-a_2)^2)}{(\frac{1}{b^2+(x-a_1)^2} + \frac{1}{b^2+(x-a_2)^2}) \cdot (b^2 + (x-a_1)^2) \cdot (b^2 + (x-a_2)^2)} \\
&= \frac{b^2 + (x-a_2)^2}{b^2 + (x-a_1)^2 + b^2 + (x-a_2)^2} \\
P(\omega_1|x) &= \frac{b^2 + (x-a_2)^2}{2b^2 + (x-a_1)^2 + (x-a_2)^2}
\end{aligned}$$

We graph $P(\omega_1|x)$ with $a_1 = 3, a_2 = 2, b = 5$, or

$$\begin{aligned}
P(\omega_1|x) &= \frac{b^2 + (x-a_2)^2}{2b^2 + (x-a_1)^2 + (x-a_2)^2} \\
&= \frac{5^2 + (x-2)^2}{2(5^2) + (x-3)^2 + (x-2)^2} \\
&= \frac{25 + x^2 + 4 - 4x}{50 + x^2 + 9 - 6x + x^2 + 4 - 4x} \\
&= \frac{x^2 - 4x + 29}{2x^2 - 10x + 63}
\end{aligned}$$

From looking at the leading terms of both the numerator and denominator, we can see that for both $x \rightarrow -\infty$ and $x \rightarrow \infty$, we simply have $\boxed{P(\omega_1|x) \rightarrow \frac{1}{2}}$. Looking at the graph itself, which we can not-so-neatly graph:



which confirms this. Now on to actually show what we wanted to show:

Setting $p(x|\omega_1) = p(x|\omega_2)$:

$$\frac{1}{\pi b} \cdot \frac{1}{1 + \left(\frac{x-a_1}{b}\right)^2} = \frac{1}{\pi b} \cdot \frac{1}{1 + \left(\frac{x-a_2}{b}\right)^2}$$

$$\frac{1}{1 + \left(\frac{x-a_1}{b}\right)^2} = \frac{1}{1 + \left(\frac{x-a_2}{b}\right)^2}$$

$$\left(\frac{x-a_1}{b}\right)^2 = \left(\frac{x-a_2}{b}\right)^2$$

$$(x-a_1)^2 = (x-a_2)^2$$

$$x^2 + a_1^2 - 2a_1x = x^2 + a_2^2 - 2a_2x$$

$$a_1^2 - 2a_1x = a_2^2 - 2a_2x$$

$$x(2a_2 - 2a_1) = a_2^2 - a_1^2$$

$$x = \frac{(a_2 - a_1)(a_2 + a_1)}{2(a_2 - a_1)}$$

$$x = \frac{a_1 + a_2}{2}$$

which is exactly what we sought to show.

(b)

Using the conditional densities in part a, and assuming equal *a priori* probabilities, show that the minimum probability of error is given by

$$P(\text{error}) = \frac{1}{2} - \frac{1}{\pi} \tan^{-1} \left| \frac{a_2 - a_1}{2b} \right|.$$

Sketch this as a function of $|(a_2 - a_1)/b|$.

We set up the error integral to determine show this minimum probability of error, choosing the decision boundary to be $\frac{a_1 + a_2}{2}$ as we previously found to be where the probabilities are equal:

$$\begin{aligned} P(\text{error}) &= P(\omega_1)P(x > \frac{a_1 + a_2}{2} | \omega_1) + P(\omega_2)P(x < \frac{a_1 + a_2}{2} | \omega_2) \\ &= \frac{1}{2} \int_{\frac{a_1 + a_2}{2}}^{\infty} p(x | \omega_1) dx + \frac{1}{2} \int_{-\infty}^{\frac{a_1 + a_2}{2}} p(x | \omega_2) dx \end{aligned}$$

The integral of the Cauchy function looks annoying, let's do it on the side:

$$\begin{aligned} p(x | \omega_i) &= \frac{1}{\pi b} \cdot \frac{1}{1 + (\frac{x - a_i}{b})^2} \\ \int p(x | \omega_i) &= F(x) = \int_{-\infty}^x \frac{1}{\pi b} \cdot \frac{1}{1 + (\frac{x - a_i}{b})^2} dx \\ &= \frac{1}{\pi} \int_{-\infty}^x \frac{1}{b} \cdot \frac{1}{1 + (\frac{x - a_i}{b})^2} dx \end{aligned}$$

Use substitution, with $x = u, v = \frac{u - a_i}{b}$, and conveniently $dv = \frac{1}{b} du$:

$$\begin{aligned} &= \frac{1}{\pi} \int_{-\infty}^{\frac{x - a_i}{b}} \frac{1}{1 + v^2} dv \\ &= \frac{1}{\pi} \int_{-\infty}^{\frac{x - a_i}{b}} \frac{1}{1 + v^2} dv \\ &= \frac{1}{\pi} \cdot \tan^{-1}(v) \Big|_{-\infty}^{\frac{x - a_i}{b}} \\ &= \frac{1}{\pi} \cdot (\tan^{-1}(\frac{x - a_i}{b}) - (-\frac{\pi}{2})) \\ F(x) &= \frac{1}{\pi} \tan^{-1}(\frac{x - a_i}{b}) + \frac{1}{2} \end{aligned}$$

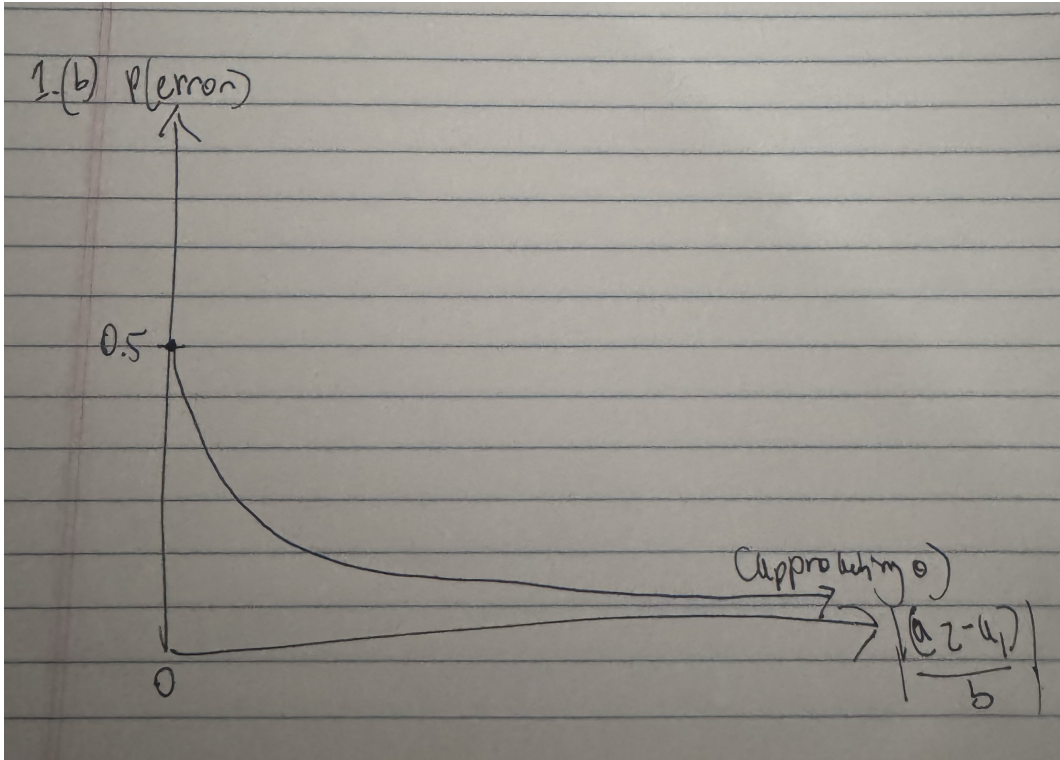
We now calculate the tail probabilities. We evaluate our CDF at the distance from the decision boundary to each class center: specifically, $\frac{a_1+a_2}{2} - a_1 = \frac{a_2-a_1}{2}$ for class ω_1 and $\frac{a_1+a_2}{2} - a_2 = \frac{a_1-a_2}{2}$ for class ω_2 . Using these distances, we find the respective tail probabilities:

$$\begin{aligned}
 - P(x > \frac{a_1+a_2}{2} | \omega_1) &= 1 - F_{a_1}(\frac{a_1+a_2}{2}) = 1 - (\frac{1}{\pi} \tan^{-1}(\frac{\frac{a_1+a_2}{2} - a_1}{b}) + \frac{1}{2}) \\
 &= 1 - (\frac{1}{\pi} \tan^{-1}(\frac{a_2-a_1}{2b}) + \frac{1}{2}) = \frac{1}{2} - \frac{1}{\pi} \tan^{-1}(\frac{a_2-a_1}{2b}) \\
 - P(x < \frac{a_1+a_2}{2} | \omega_2) &= F_{a_2}(\frac{a_1+a_2}{2}) = \frac{1}{\pi} \tan^{-1}(\frac{\frac{a_1+a_2}{2} - a_2}{b}) + \frac{1}{2} = \frac{1}{2} - \frac{1}{\pi} \tan^{-1} \frac{a_2-a_1}{2b}
 \end{aligned}$$

Finally, plugging all of this back into our original error equation:

$$\begin{aligned}
 P(\text{error}) &= \frac{1}{2} \int_{\frac{a_1+a_2}{2}}^{\infty} p(x|\omega_1) dx + \frac{1}{2} \int_{-\infty}^{\frac{a_1+a_2}{2}} p(x|\omega_2) dx \\
 &= \frac{1}{2} \cdot (\frac{1}{2} - \frac{1}{\pi} \tan^{-1}(\frac{a_2-a_1}{2b})) + \frac{1}{2} \cdot (\frac{1}{2} - \frac{1}{\pi} \tan^{-1}(\frac{a_2-a_1}{2b})) \\
 P(\text{error}) &= \frac{1}{2} - \frac{1}{\pi} \tan^{-1}(\frac{|a_2-a_1|}{2b})
 \end{aligned}$$

exactly as we sought to find. Sketching this as a function of $|(a_2 - a_1)/b|$:



2.

The Poisson distribution for a discrete $k, k = 0, 1, 2, \dots$ and real parameter λ is

$$P(k|\lambda) = e^{-\lambda} \frac{\lambda^k}{k!}.$$

(a)

Find the mean of k .

By definition, we have that

$$\begin{aligned} E[k] &= \sum_{k=0}^{\infty} k \cdot P(k|\lambda) \\ &= \sum_{k=0}^{\infty} k \cdot e^{-\lambda} \frac{\lambda^k}{k!} \\ &= 0 + \sum_{k=1}^{\infty} \lambda^k e^{-\lambda} \frac{k}{k!} \\ &= e^{-\lambda} \sum_{k=1}^{\infty} \lambda^k \frac{1}{(k-1)!} \end{aligned}$$

Simplifying by substituting $j = k - 1$:

$$\begin{aligned} &= e^{-\lambda} \sum_{j=0}^{\infty} \lambda^{j+1} \frac{1}{j!} \\ &= \lambda e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} \\ &= \lambda e^{-\lambda} \cdot e^{\lambda} \end{aligned}$$

$$\boxed{E[k] = \lambda}$$

(b)

Find the variance of k .

We need to essentially find $E[k^2]$ here. We can quickly ascertain that directly substituting k^2 will make this hard to simplify and cancel out the k terms in both the numerator and factorial denominator, so we can instead solve for $E[k(k-1)]$:

$$\begin{aligned}
E[k(k-1)] &= \sum_{k=0}^{\infty} k(k-1) \cdot e^{-\lambda} \frac{\lambda^k}{k!} \\
&= \sum_{k=0}^{\infty} e^{-\lambda} \frac{(k-1) \cdot \lambda^k}{(k-1)!} \\
&= e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{(k-2)!} \\
&= 0 + 0 + e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^k}{(k-2)!}
\end{aligned}$$

Simplifying by substituting $j = k - 2$:

$$\begin{aligned}
&= e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^{j+2}}{j!} \\
&= \lambda^2 e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} \\
E[k(k-1)] &= \lambda^2 e^{-\lambda} e^{\lambda} = \lambda^2 \\
E[k^2 - k] &= \lambda^2 \\
E[k^2] - E[k] &= \lambda^2 \\
E[k^2] &= \lambda^2 + E[k] = \lambda^2 + \lambda
\end{aligned}$$

We can now solve for variance:

$$\begin{aligned}
\text{Var}(k) &= E[k^2] - E[k]^2 \\
&= \lambda^2 + \lambda - \lambda^2 \\
\boxed{\text{Var}(k) &= \lambda}
\end{aligned}$$

(c)

Find the mode of k .

The mode is the value of k with the largest probability of occurring. Since k is an integer value, it must be an integer value. We can determine this by observing the nature in which the sequence continues element by element:

$$P(k|\lambda) =$$
$$P(k+1|\lambda) = e^{-\lambda} \frac{\lambda^{k+1}}{(k+1)!}$$

We see that

$$\frac{P(k|\lambda)}{P(k+1|\lambda)} = \frac{e^{-\lambda} \frac{\lambda^k}{k!}}{e^{-\lambda} \frac{\lambda^{k+1}}{(k+1)!}}$$
$$= \frac{\lambda^k \cdot (k+1)!}{\lambda^{k+1} \cdot k!}$$
$$= \frac{k+1}{\lambda}$$

So essentially, with the way this is set up, we want to find when the current probability is bigger than the next probability, since we know the Poisson distribution is parabolically shaped a maximum point after which the probability begins decreasing. We have a few different cases:

- If $\frac{k+1}{\lambda} > 1$, then probabilities are going down.
- If $\frac{k+1}{\lambda} < 1$, the probabilities are going up.
- If $\frac{k+1}{\lambda} = 1$, then $k = \lambda - 1$.

From looking at this, we can see that the highest probability or mode happens around $k = \lambda - 1$. However, since it is possible for $k = \lambda - 1$, in this case in which λ is an integer, we need to differentiate two cases, since both $\boxed{\lambda - 1}$ and $\boxed{\lambda}$ have equal probability and are both integers. In the case in which λ is not an integer, then since k must be an integer, we just take $\boxed{\lambda \text{ floored to the nearest integer}}$.

(d)

Assume two categories C_1 and C_2 , equally probably *a priori*, distributed with Poisson distributions and $\lambda_1 > \lambda_2$. What is the Bayes classification decision?

We have $P(C_1) = P(C_2) = 0.5$, and $P(x|C_1) = e^{-\lambda_1} \frac{\lambda_1^x}{x!}$ and $P(x|C_2) = e^{-\lambda_2} \frac{\lambda_2^x}{x!}$. By definition, we decide the category based on the probability comparison, since the priors are equal, i.e. we would choose C_1 if $P(C_1) > P(C_2)$, but as they are equal, we see if $P(x|C_1) > P(x|C_2)$:

$$\begin{aligned} e^{-\lambda_1} \frac{\lambda_1^x}{x!} &> e^{-\lambda_2} \frac{\lambda_2^x}{x!} \\ e^{-\lambda_1} \lambda_1^x &> e^{-\lambda_2} \lambda_2^x \\ -\lambda_1 + x \ln(\lambda_1) &> -\lambda_2 + x \ln(\lambda_2) \\ x \ln(\lambda_1) - x \ln(\lambda_2) &> -\lambda_2 + \lambda_1 \\ x(\ln(\lambda_1) - \ln(\lambda_2)) &> \lambda_1 - \lambda_2 \\ x &> \frac{\lambda_1 - \lambda_2}{\ln(\lambda_1) - \ln(\lambda_2)} \end{aligned}$$

Therefore, the decision rule is as follows:

Choose C_1 if $x > \frac{\lambda_1 - \lambda_2}{\ln(\lambda_1) - \ln(\lambda_2)}$; Otherwise, choose C_2 .

(e)

What is the Bayes error rate?

Logically, the Bayes error rate is the probability of making an incorrect classification; that is, either we choose C_2 when C_1 was the right category, or we choose C_1 when C_2 was the right category. Using our priors with $P(C_1) = P(C_2) = 0.5$, and letting the threshold of classification to be $y = \frac{\lambda_1 - \lambda_2}{\ln(\lambda_1) - \ln(\lambda_2)}$, with the understanding that we classify into the categories based on being above or below this threshold, we can express $P(\text{error})$ to be

$$P(\text{error}) = P(C_1) \cdot \sum_{k=0}^{\lfloor y \rfloor} P(k|C_1) + P(C_2) \cdot \sum_{k=\lceil y \rceil}^{\infty} P(k|C_2)$$

$$P(\text{error}) = \frac{1}{2} \cdot \sum_{k=0}^{\lfloor y \rfloor} e^{-\lambda_1} \frac{\lambda_1^k}{k!} + \frac{1}{2} \cdot \sum_{k=\lceil y \rceil}^{\infty} e^{-\lambda_2} \frac{\lambda_2^k}{k!}$$

$$P(\text{error}) = \frac{\sum_{k=0}^{\lfloor y \rfloor} e^{-\lambda_1} \frac{\lambda_1^k}{k!} + \sum_{k=\lceil y \rceil}^{\infty} e^{-\lambda_2} \frac{\lambda_2^k}{k!}}{2}$$

3.

Let $p(x|\omega_i) \sim N(\mu_i, \sigma^2 I)$ for a two-category k -dimensional problem with $P(\omega_1) = P(\omega_2) = \frac{1}{2}$.

(a)

Find P_e , the minimum probability of error.

To apply the Bayes Decision Rule, we decide ω_1 if the likelihood ratio is greater than our relevant threshold. Here, with $P(\omega_1) = P(\omega_2) = \frac{1}{2}$, we basically are directly comparing the probabilities. We know that we are essentially choosing ω_1 based on the truthiness of

$$p(x|\omega_1) > p(x|\omega_2)$$

Substituting in the PDF for a k -dimensional multivariate normal distribution with $\Sigma = \sigma^2 I$, i.e. $p(x|\omega_i) = \frac{1}{(2\pi\sigma^2)^{\frac{k}{2}}} e^{-\frac{\|x-\mu_i\|^2}{2\sigma^2}}$:

$$\begin{aligned} \frac{1}{(2\pi\sigma^2)^{\frac{k}{2}}} e^{-\frac{\|x-\mu_2\|^2}{2\sigma^2}} &> \frac{1}{(2\pi\sigma^2)^{\frac{k}{2}}} e^{-\frac{\|x-\mu_1\|^2}{2\sigma^2}} \\ e^{-\frac{\|x-\mu_1\|^2}{2\sigma^2}} &> e^{-\frac{\|x-\mu_2\|^2}{2\sigma^2}} \end{aligned}$$

Take the log-likelihood to make it easier to interpret:

$$\begin{aligned} -\frac{\|x-\mu_1\|^2}{2\sigma^2} &> -\frac{\|x-\mu_2\|^2}{2\sigma^2} \\ \|x-\mu_1\|^2 &< \|x-\mu_2\|^2 \end{aligned}$$

Expanding by definition of the norm function and solving:

$$(x - \mu_1)^T (x - \mu_1) < (x - \mu_2)^T (x - \mu_2)$$

$$x^T x - 2\mu_1^T x + \mu_1^T \mu_1 < x^T x - 2\mu_2^T x + \mu_2^T \mu_2$$

$$2\mu_2^T x - 2\mu_1^T x < \mu_2^T \mu_2 - \mu_1^T \mu_1$$

$$2(\mu_2 - \mu_1)^T x < \|\mu_2\|^2 - \|\mu_1\|^2$$

$$(\mu_2 - \mu_1)^T x < \frac{1}{2}(\|\mu_2\|^2 - \|\mu_1\|^2)$$

We now have our discriminant function which is the left-hand side, with the threshold on the right-hand side. We now have distribution $g(x) = (\mu_2 - \mu_1)^T x$ which is a normal random variable with mean $\mu_g = E[g(x)|\omega_1] = (\mu_2 - \mu_1)^T \mu_1$, and variance $\sigma_g^2 = \text{Var}(g(x)|\omega_1) = (\mu_2 - \mu_1)^T (\sigma^2 I) (\mu_2 - \mu_1) = \sigma^2 \|\mu_2 - \mu_1\|^2$. Finally, we now compute the minimum probability of error:

$$P(\text{error}|\omega_1) = P(g(x) \geq \text{threshold}|\omega_1) = \Phi\left(\frac{\mu_g - \text{threshold}}{\sigma_g}\right)$$

Now substituting and simplifying:

$$\begin{aligned} &= \Phi\left(\frac{(\mu_2 - \mu_1)^T \mu_1 - \frac{1}{2}(\|\mu_2\|^2 - \|\mu_1\|^2)}{\sigma \|\mu_2 - \mu_1\|}\right) \\ &= \Phi\left(\frac{\mu_2^T \mu_1 - \mu_1^T \mu_1 - \frac{1}{2}\|\mu_2\|^2 + \frac{1}{2}\|\mu_1\|^2}{\sigma \|\mu_2 - \mu_1\|}\right) \\ &= \Phi\left(\frac{\mu_2^T \mu_1 - \|\mu_1\|^2 - \frac{1}{2}\|\mu_2\|^2 + \frac{1}{2}\|\mu_1\|^2}{\sigma \|\mu_2 - \mu_1\|}\right) \\ &= \Phi\left(\frac{-\frac{1}{2}(\|\mu_1\|^2 + \|\mu_2\|^2 - 2\mu_2^T \mu_1)}{\sigma \|\mu_2 - \mu_1\|}\right) \\ &= \Phi\left(\frac{-\frac{1}{2}\|\mu_2 - \mu_1\|^2}{\sigma \|\mu_2 - \mu_1\|}\right) \\ &\quad \boxed{P_e = \Phi\left(-\frac{\|\mu_2 - \mu_1\|}{2\sigma}\right)} \end{aligned}$$

(b)

Let $\mu_1 = 0$ and $\mu_2 = (m_1, \dots, m_k)^T \neq 0$. Show that $P_e \rightarrow 0$ as the dimension k approaches infinity. Assume that $\sum_{k=1}^{\infty} m_k^2 \rightarrow \infty$.

The squared difference between the means (which we determined what we needed for doing the decisionmaking) would be

$$\|\mu_2 - \mu_1\|^2 = \sum_{j=1}^k m_j^2$$

and therefore

$$\|\mu_2 - \mu_1\| = \sqrt{\sum_{j=1}^k m_j^2}$$

So we therefore can substitute accordingly to find that

$$P_e = \Phi\left(-\frac{\sqrt{\sum_{j=1}^k m_j^2}}{2\sigma}\right)$$

We are given in the problem the assumption that $\sum_{k=1}^{\infty} m_k^2 \rightarrow \infty$, so substituting we see that

$$-\frac{\sqrt{\sum_{j=1}^k m_j^2}}{2\sigma} \rightarrow -\frac{\sqrt{\infty}}{2\sigma} \rightarrow -\infty$$

and substituting once again, we find that

$$P_e = \Phi(-\infty) \rightarrow 0$$

exactly as we sought to find.

4.

Under the assumption that $\lambda_{21} > \lambda_{11}$ and $\lambda_{12} > \lambda_{22}$, show that the general minimum risk discriminant function for a classifier with independent binary features is given by $g(x) = w^T x + w_0$. What are w and w_0 ?